

# Analysis of Two-Mode Networks Using R

## 1. Introduction

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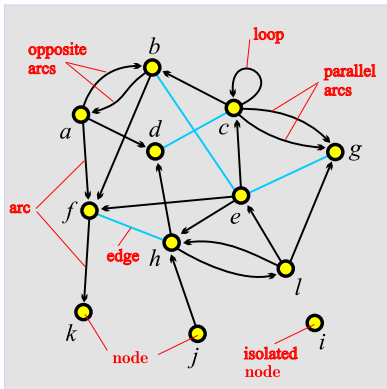
- 1 Networks
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**Current version of slides (August 13, 2026 at 13 : 25):** [slides PDF](#)

<https://github.com/bavla/Nets/blob/master/ws/ws26.md>



unit, actor – *node*, vertex  
tie, line – *link*, edge, arc

*arc* = directed link, (*a*, *d*)  
*a* is the *initial* node,  
*d* is the *terminal* node.

*edge* = undirected link,  
(*c*: *d*)  
*c* and *d* are *end* nodes.

# Display of properties – school (Moody)

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Networks

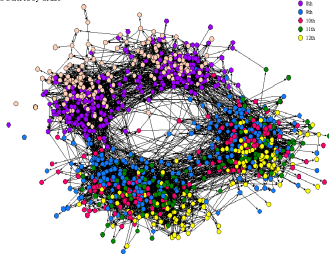
Properties

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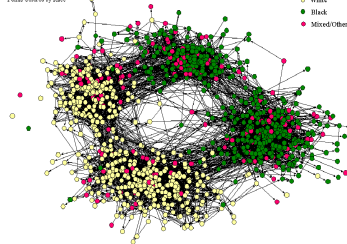
The Social Structure of "Countryside" School District

Points Colored by Grade



The Social Structure of "Countryside" School District

Points Colored by Race

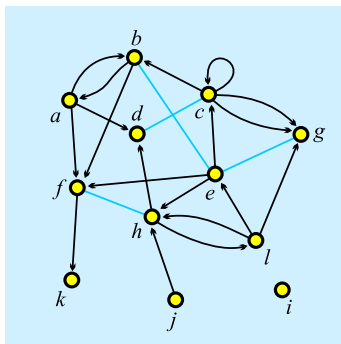


**Besides the graph, we need data to understand the network!**

## Network = Graph + Data

A *network*  $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$  consists of:

- a *graph*  $\mathcal{G} = (\mathcal{V}, \mathcal{L})$ , where  $\mathcal{V}$  is the set of nodes,  $\mathcal{A}$  is the set of arcs,  $\mathcal{E}$  is the set of edges, and  $\mathcal{L} = \mathcal{E} \cup \mathcal{A}$  is the set of links.  
 $n = |\mathcal{V}|$ ,  $m = |\mathcal{L}|$
- $\mathcal{P}$  *node value functions* / properties:  $p: \mathcal{V} \rightarrow A$
- $\mathcal{W}$  *link value functions* / weights:  $w: \mathcal{L} \rightarrow B$



$$\mathcal{V} = \{a, b, c, d, e, f, g, h, i, j, k, l\}$$

$$\mathcal{A} = \{(a, b), (a, d), (a, f), (b, a), (b, f), (c, b), (c, c), (c, g)_1, (c, g)_2, (e, c), (e, f), (e, h), (f, k), (h, d), (h, l), (j, h), (l, e), (l, g), (l, h)\}$$

$$\mathcal{E} = \{(b: e), (c: d), (e: g), (f: h)\}$$

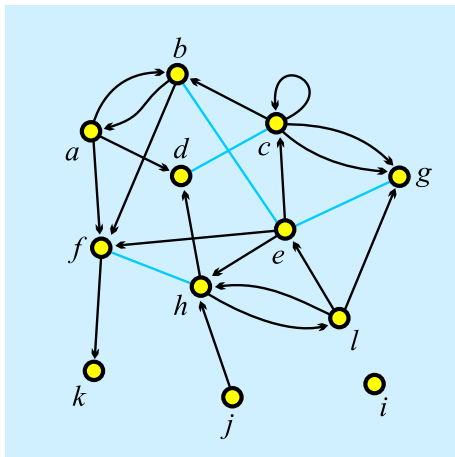
$$\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{E})$$

$$\mathcal{L} = \mathcal{A} \cup \mathcal{E}$$

$\mathcal{A} = \emptyset$  – **undirected** graph;  $\mathcal{E} = \emptyset$  – **directed** graph.

The R package igraph **doesn't support** the mixing of arcs and edges. A graph is either directed or undirected.

In most applications, an edge ( $u: v$ ) can be replaced by a pair of opposite arcs ( $u, v$ ) and ( $v, u$ ).

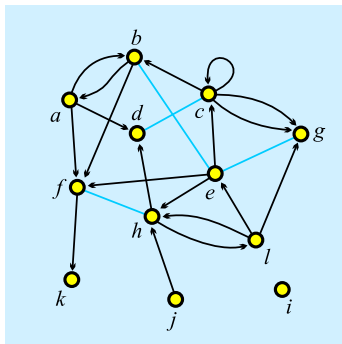


```
*Vertices 12
1 "a" 0.1020 0.3226
2 "b" 0.2860 0.0876
3 "c" 0.5322 0.2304
4 "d" 0.3259 0.3917
5 "e" 0.5543 0.4770
6 "f" 0.1552 0.6406
7 "g" 0.8293 0.3249
8 "h" 0.4479 0.6866
9 "i" 0.8204 0.8203
10 "j" 0.4789 0.9055
11 "k" 0.1175 0.9032
12 "l" 0.7095 0.6475

*Arcs
1 2
2 1
1 4
1 6
2 6
3 2
3 3
3 7
3 7
5 3
5 6
5 8
6 11
8 4
10 8
12 5
12 7
8 12
12 8

*Edges
2 5
3 4
5 7
6 8
```

factorization !!!



|   | a | b | c | d | e | f | g | h | i | j | k | l |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| a | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| b | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| c | 0 | 1 | 1 | 1 | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 0 |
| d | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| e | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| f | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| g | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| h | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| i | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| j | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| k | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| l | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |

The graph  $G$  is *simple* if in the corresponding matrix all entries are 0 or 1.

Matrix representation provides a link to the tools of linear algebra.



# Networks in igraph

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The network  $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$  can be described by two tables  $(\mathcal{V}, \mathcal{P})$  and  $(\mathcal{L}, \mathcal{W})$ .

GraphSet.net, nodes.csv, links.csv

```
> nWdir <- paste0("https://raw.githubusercontent.com/",  
+ "bavla/Nets/refs/heads/master/netsWeight/")  
> (V <- read.csv(paste0(nWdir, "data/nodes.csv"), sep=""))  
> (L <- read.csv(paste0(nWdir, "data/links.csv"), sep=""))  
> (N <- graph_from_data_frame(L, directed=TRUE, vertices=V))  
> V(N)$name <- V(N)$label  
> N <- delete_vertex_attr(N, "label")  
> E(N)$weight <- sample(1:7, ecount(N), replace=TRUE)  
> N  
> plot(N, edge.width=E(N)$weight)  
> N$name <- "Network from data frame example"  
> N$by <- "Vladimir Batagelj"  
> N$date <- date()  
> N  
> saveRDS(N, file="igraphDF.rds")  
  
> graph_attr(N)  
> (nodes <- as_data_frame(N, what="vertices"))  
> (links <- as_data_frame(N, what="edges"))
```

Besides ordinary (directed, undirected, mixed) networks, some special types of networks are also used:

- *2-mode networks*, bipartite (valued) graphs – networks between two disjoint sets of nodes.
- *multi-relational networks*.
- *linked networks* and *collections of networks*.
- *multilevel networks*.
- *temporal networks*, dynamic graphs – networks changing over time.
- specialized networks: representation of genealogies as *p-graphs*; *Petri's nets*, molecular graphs, etc.

Network (input) file formats should provide the means of expressing all of these types of networks. All interesting data should be recorded (respecting privacy).

In a *two-mode* network  $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, \mathcal{P}, \mathcal{W})$  the set of nodes consists of two disjoint sets of nodes  $\mathcal{U}$  and  $\mathcal{V}$ , and all the links from  $\mathcal{L}$  have one end node in  $\mathcal{U}$  and the other in  $\mathcal{V}$ .

A *multi-relational network*  $\mathcal{N} = (\mathcal{V}, (\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_k), \mathcal{P}, \mathcal{W})$  contains different relations  $\mathcal{L}_i$  (sets of links) over the same set of nodes. Also, the weights from  $\mathcal{W}$  are defined on different relations or their union.

In a *linked* or *multimodal* network  $\mathcal{N} = ((\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_j), (\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_k), \mathcal{P}, \mathcal{W})$  the set of nodes  $\mathcal{V}$  is partitioned into subsets (*modes*)  $\mathcal{V}_i$ ,  $\mathcal{L}_s \subseteq \mathcal{V}_p \times \mathcal{V}_q$ , and properties and weights are usually partial functions.

A set of networks  $\{\mathcal{N}_1, \mathcal{N}_2, \dots, \mathcal{N}_k\}$  in which each network shares a (sub)set of nodes with some other network is called a *collection* of networks.

A linked network can be transformed into a collection of networks and vice versa.

Bibliographical information is usually represented as a collection of bibliographical networks  $\{\mathbf{Ci}, \mathbf{WA}, \mathbf{WK}, \mathbf{WC}, \mathbf{WI}, \dots\}$  (W – works, A – authors, K – keywords, C – countries, I – institutions) [4].

Another example of multimodal multirelational networks are knowledge graphs. They can have a very diverse structure (a large number of types of units (modes) and predicates (relations)), which allows for a fairly accurate description of facts from a selected field and solving problems about it.



# netsJSON

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For archiving networks, we can use the JSON format supported by the package `netsWeight` [6, 7].

```
{
  "netsJSON": "basic",
  "info": {
    "org":1, "nNodes":n, "nArcs":mA, "nEdges":mE,
    "simple":TF, "directed":TF, "multirel":TF, "mode":m,
    "network":fName, "title":title,
    "time": { "Tmin":tm, "Tmax":tM, "Tlabs": {labs} },
    "meta": [events], ...
  },
  "nodes": [
    { "id":nodeId, "lab":label, "x":x, "y":y, ...},
    ***
  ],
  "links": [
    { "type":arc/edge, "n1":nodeID1, "n2":nodeID2, "rel":r, ...},
    ***
  ],
  "data": {
    "data1":description1,
    ***
  }
}
```



# netsJSON

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```
> saveRDS(AW, file="AW.rds")
> AW1 <- readRDS(file="AW.rds")
> AW1
> write_graph_netsJSON(AW, file="AW.json")
> AW2 <- netsJSON_to_graph(fromJSON("AW.json"), directed=TRUE)
> AW2
> AW3 <- netsJSON_to_graph(fromJSON("AW.json"), directed=FALSE)
> AW3
> V(AW3) [[ ]]
> E(AW3) [[ ]]
> graph_attr(AW3)
```

We can combine individual networks into a collection

```
> Bib <- list(netsJSON="Collection",
+ info=list(date=date(), by="VB"), nets=list(AW=AW, WK=WK))
> str(Bib)
> saveRDS(Bib, file="BibNets.rds")
```

*Properties* of nodes  $\mathcal{P}$  and links  $\mathcal{W}$  (weights) can be measured on different scales: *numerical* (age, weight, number of contacts), *ordinal* (level of education), and *nominal/categorical* (citizenship, sex, type of contact). They can be *input* as data or *computed* from the network.

Recently, properties with *structured* values are also considered: intervals, lists of keywords, event sequences, distributions, temporal quantities, etc.

We also have to specify how to compute using the values of each property. Usually, for a numerical property  $p : \mathcal{V} \rightarrow \mathbb{R}$ , the real number addition  $+$  is used. In a general case  $p : \mathcal{V} \rightarrow S$ , we assume that  $(S, +, 0)$  is an *Abelian monoid*; and in the case of a weight  $w : \mathcal{L} \rightarrow S$ ,  $(S, +, \cdot, 0, 1)$  is a *semiring* [1].

# Example of semiring

To construct a semiring corresponding to the balance problem in signed networks ( $w : \mathcal{L} \rightarrow \{n, p\}$ ,  $n$  – negative,  $p$  – positive) we take the set  $S = \{0, n, p, a\}$  with four elements (Doreian, 1970; Harary et al., 1965):

- $0$  – no walk;
- $n$  – all walks are negative;
- $p$  – all walks are positive;
- $a$  – at least one positive and at least one negative walk.

Balance Semiring

| $+$ | $0$ | $n$ | $p$ | $a$ | $\cdot$ | $0$ | $n$ | $p$ | $a$ | $x$ | $x^*$ |
|-----|-----|-----|-----|-----|---------|-----|-----|-----|-----|-----|-------|
| $0$ | $0$ | $n$ | $p$ | $a$ | $0$     | $0$ | $0$ | $0$ | $0$ | $0$ | $p$   |
| $n$ | $n$ | $n$ | $a$ | $a$ | $n$     | $0$ | $p$ | $n$ | $a$ | $n$ | $a$   |
| $p$ | $p$ | $a$ | $p$ | $a$ | $p$     | $0$ | $n$ | $p$ | $a$ | $p$ | $p$   |
| $a$ | $a$ | $a$ | $a$ | $a$ | $a$     | $0$ | $a$ | $a$ | $a$ | $a$ | $a$   |



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Traditional measurements can be represented (using factorization) as a numerical vector  $\mathbf{v} = [v_i]$ .

**Clustering** – partition of elements (nodes/links) – *nominal* or *ordinal* data about elements

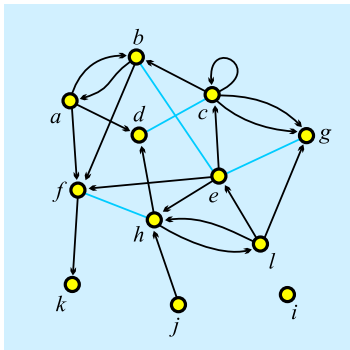
$v_i \in \mathbb{N}$  : element  $i$  belongs to the cluster/group  $v_i$ ;

**Vector** – *numeric* data about elements

$v_i \in \mathbb{R}$  : the property has value  $v_i$  on element  $i$ ;

**Permutation** – *ordering* of elements

$v_i \in \mathbb{N}$  : element  $i$  is at the  $v_i$ -th position.



*degree* of node  $v$ ,  $\deg(v)$  = number of links with  $v$  as an endnode – a *star*;

*indegree* of node  $v$ ,  $\text{ideg}(v)$  = number of links with  $v$  as a terminal node – an *instar*. An edge endnode is both initial and terminal;

*outdegree* of node  $v$ ,  $\text{odeg}(v)$  = number of links with  $v$  as an initial node – an *outstar*.

For a graph,

*initial* node  $v \Leftrightarrow \text{ideg}(v) = 0$

*terminal* node  $v \Leftrightarrow \text{odeg}(v) = 0$

$$n = 12, m = 23, \text{ideg}(e) = 3, \text{odeg}(e) = 5, \deg(e) = 6$$

$$\sum_{v \in \mathcal{V}} \text{ideg}(v) = \sum_{v \in \mathcal{V}} \text{odeg}(v) = |\mathcal{A}| + 2|\mathcal{E}| - |\mathcal{E}_0|, \sum_{v \in \mathcal{V}} \deg(v) = 2|\mathcal{L}| - |\mathcal{L}_0|$$

## *weighted in/out-degree*

$$\text{wd}(v) = \sum_{e \in \text{star}(v)} w(e)$$

$$\text{wid}(v) = \sum_{e \in \text{istar}(v)} w(e)$$

$$\text{wod}(v) = \sum_{e \in \text{ostar}(v)} w(e)$$

$$\text{wto}(v, u) = \sum_{e \in \text{ostar}(v) \cap \text{istar}(u)} w(e)$$



# Neighbors

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$N(v)$  is the set of *neighbors* of node  $v$  – other endnodes of links containing the node  $v$ .

$iN(v)$  is the set of *in-neighbors* of node  $v$  – other nodes from which a link leads to the node  $v$ .

$oN(v)$  is the set of *out-neighbors* of node  $v$  – other nodes to which a link leads from the node  $v$ .

# Important nodes and links in network

To identify important/interesting elements (nodes, links) in a network, we often try to express our intuition about important/interesting elements using an appropriate measure (index, weight) following the scheme

*The larger is the measured value of an element,  
the more important/interesting this element is*

Too often, in the analysis of networks, researchers uncritically pick some measures from the literature. For a formal approach, see [Roberts](#).

The (importance) measure can be obtained as input data – an *observed* property, or computed from the network description – a *structural* property.

An interesting question is studying associations among structural and observed properties. Can an observed property be explained with some structural property/ies?



# ... Important nodes and links in network

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The most important distinction between different node *indices* is based on the view/decision of whether the network is considered directed or undirected – closeness, betweenness, clustering, eigen-vector, ...

Using node or link cuts, islands, cores, skeletons (spanning tree, Pathfinder, k-neighbors), community detection, hubs and authorities, etc., we can identify the most active subnetworks. We can also apply clustering and blockmodeling methods [3].

The **node-cut** of a network  $\mathcal{N} = (\mathcal{V}, \mathcal{L}, p)$ ,  $p : \mathcal{V} \rightarrow \mathbb{R}$ , at selected level  $t$  is a subnetwork  $\mathcal{N}(t) = (\mathcal{V}', \mathcal{L}(\mathcal{V}'), p)$ , determined by the set

$$\mathcal{V}' = \{v \in \mathcal{V} : p(v) \geq t\}$$

and  $\mathcal{L}(\mathcal{V}')$  is the set of links from  $\mathcal{L}$  that have both endnodes in  $\mathcal{V}'$ .

The **link-cut** of a network  $\mathcal{N} = (\mathcal{V}, \mathcal{L}, w)$ ,  $w : \mathcal{L} \rightarrow \mathbb{R}$ , at selected level  $t$  is a subnetwork  $\mathcal{N}(t) = (\mathcal{V}(\mathcal{L}'), \mathcal{L}', w)$ , determined by the set

$$\mathcal{L}' = \{e \in \mathcal{L} : w(e) \geq t\}$$

and  $\mathcal{V}(\mathcal{L}')$  is the set of all endnodes of the links from  $\mathcal{L}'$ .



# netsWeight

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We started to develop an R package `netsWeight` over `igraph`, extending its capabilities with some tools for analysis of weighted networks. The package `netsWeight` contains functions

`interlinks(N, atn, c1, c2, col1="red", col2="blue")` – extracts from a given network  $N$ , a subnetwork determined by node clusters  $C_1$  and  $C_2$ , and links between them.  $C_i$  is the set of nodes having for the attribute with name  $atn$  the value  $c_i$ . Its nodes have color  $col_i$ .

`node_cut(N, atn, t)` – node cut in  $N$  for the node property  $atn$  at threshold  $t$ .

`link_cut(N, atn, t)` – link cut in  $N$  for the link property  $atn$  at threshold  $t$ .

The threshold value  $t$  is determined based on the distribution of values of weight  $w$  or property  $p$ . Usually, we are interested in cuts that are not too large, but also not trivial.

# Weighted network matrix

A two mode network  $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, w)$  can be represented with the corresponding **matrix**  $\mathbf{N} = [n[u, v]]_{u \in \mathcal{U}, v \in \mathcal{V}}$  defined as

$$n[u, v] = \begin{cases} \text{wto}(u, v) & (u, v) \in \mathcal{L} \\ \square & \text{otherwise} \end{cases}$$

We represent the number 0 with two symbols, 0 (weight 0) and  $\square$  (no link) where  $\square = 0$  with rules  $\square + a = a$  and  $\square \cdot a = \square$ .

In the case  $\mathcal{U} = \mathcal{V}$  we obtain a square matrix of an ordinary network.

*When collecting the network data, consider providing as many properties as possible.*

# Binarization and transposition

A *binarization*  $\text{bin}(\mathbf{N})$  of the network matrix  $\mathbf{N}$

$$\text{bin}(\mathbf{N})[u, v] = \begin{cases} 1 & n[u, v] \neq \square \\ \square & n[u, v] = \square \end{cases}$$

The *transpose*  $\mathbf{N}^T$  of matrix  $\mathbf{N}$  is a matrix on  $\mathcal{V} \times \mathcal{U}$  with entries  $n^T[u, v] = n[v, u]$ .

$$\mathbf{VU} = \mathbf{UV}^T \text{ and } uv^T[u, v] = vu[u, v].$$

The corresponding network  $\tilde{\mathcal{N}} = ((\mathcal{V}, \mathcal{U}), \tilde{\mathcal{L}}, \tilde{n})$  is called the *reversed* network. The direction of its arcs is reversed  $\tilde{\mathcal{L}} = \{(v, u) : (u, v) \in \mathcal{L}\}$ , but the weights remain unchanged  $\tilde{n}(v, u) = n(u, v)$  for  $(u, v) \in \mathcal{L}$ .

$$\text{wod}_{\tilde{\mathcal{N}}}(v) = \text{wid}_{\mathcal{N}}(v), \dots$$



Batagelj, Vladimir : Semirings for social networks analysis, The Journal of Mathematical Sociology, 19(1994)1, 53-68.



Batagelj, V, Mrvar, A: Density based approaches to network analysis; Analysis of Reuters terror news network. Workshop on link analysis for detecting complex behavior (LinkKDD2003). [cmu/dunja](#).



Batagelj, V., Doreian, P., Ferligoj, A., Kejžar, N.: Understanding Large Temporal Networks and Spatial Networks: Exploration, Pattern Searching, Visualization and Network Evolution. Wiley, October 2014.



Batagelj, V. On fractional approach to analysis of linked networks. [Scientometrics](#) 123, 621–633 (2020).  
<https://doi.org/10.1007/s11192-020-03383-y>



Batagelj, V, Cerinšek, M: On bibliographic networks. [Scientometrics](#) 96 (2013) 3, 845-864.



Batagelj, Vladimir. Analysis of the Southern women network using fractional approach. Social Networks 68 (2022): 229-236.



Batagelj, V: On fractional approach to analysis of linked networks. Scientometrics 123 (2020) 2: 621-633



Batagelj, Vladimir, Pisanski, Tomaž, Savnik, Iztok, Slavec, Ana, Bašić, Nino: Towards a Format for Describing Networks. Information Society 2025, 6–10 October 2025, Ljubljana, Slovenia, SiKDD, 102–105.



Zachary, P Neal et al.: Recommendations for sharing network data and materials. Network Science, 12(2024)4, 404–417.



Batagelj, V, Maltseva, D: Temporal bibliographic networks. Journal of Informetrics, Volume 14, Issue 1, February 2020, 101006.



Maltseva, D, Batagelj, V: Social network analysis as a field of invasions: bibliographic approach to study SNA development. Scientometrics, 121(2019)2, 1085-1128



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<https://github.com/bavla/Nets> ; <https://github.com/bavla/TQ>



<https://github.com/bavla/NormNet/tree/main/TwoMode>